CS215 : Introduction to software Engineering Assignment 1

FACTUALY OF COMPUTER AND ARTIFICIAL INTELLIGENCE, CAIRO UNIVERSITY

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Software Engineering Assignment 2 — Task 2 Draft

BudgetWise — Personal Budgeting App

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Task 2 — Software Design Specification (Draft)

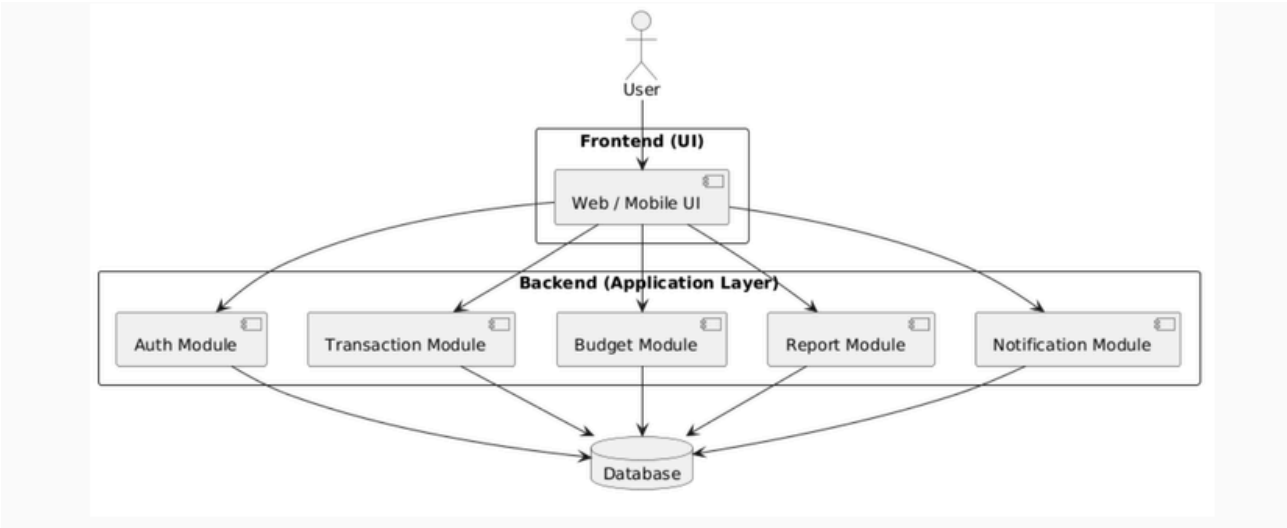
Based on the BudgetWise SRS — Personal Budgeting Software. This draft Software Design Specification (SDS) covers the system architecture, class diagram, class descriptions, and seven sequence diagrams for the most important user stories. All diagrams are generated using PlantUML. The PlantUML source code for each diagram is provided at the end of this document.

2.1 System Architecture

The BudgetWise system follows a 3-Tier MVC (Model-View-Controller) architecture. This pattern separates the application into three interconnected layers, ensuring modularity, testability, and maintainability.

Layer	Components	Responsibility
Presentation (View)	LoginScreen, DashboardScreen, TransactionScreen, BudgetScreen, GoalScreen, ReportScreen	Renders UI; captures user input; displays data from Controller
Business Logic (Controller)	AuthController, TransactionController, BudgetController, GoalController, ReportController	Processes user actions; applies business rules; coordinates between View and Model
Data / Persistence (Model + DB)	User, Transaction, Budget, Goal, Category, DatabaseManager	Stores domain data; persists to SQLite/PostgreSQL; retrieves records on demand

Architecture Diagram



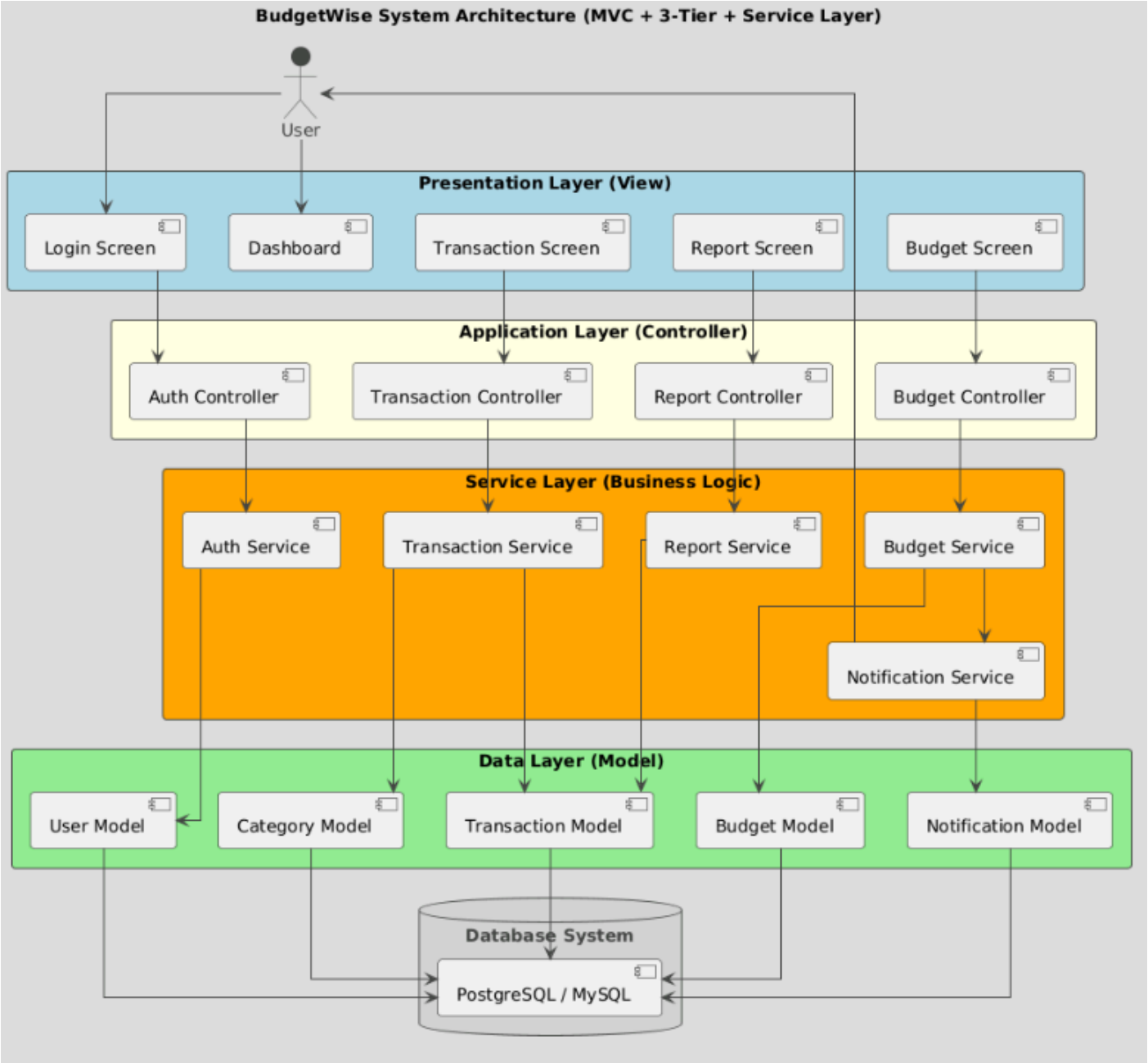


Figure A1: BudgetWise 3-Tier MVC

Architecture Diagram PlantUML Code: See

Section 2.5 — Code Block A1

2.2 Class Diagram

The class diagram below captures all domain, boundary, and control classes of the BudgetWise system, organised into three packages corresponding to the MVC layers.

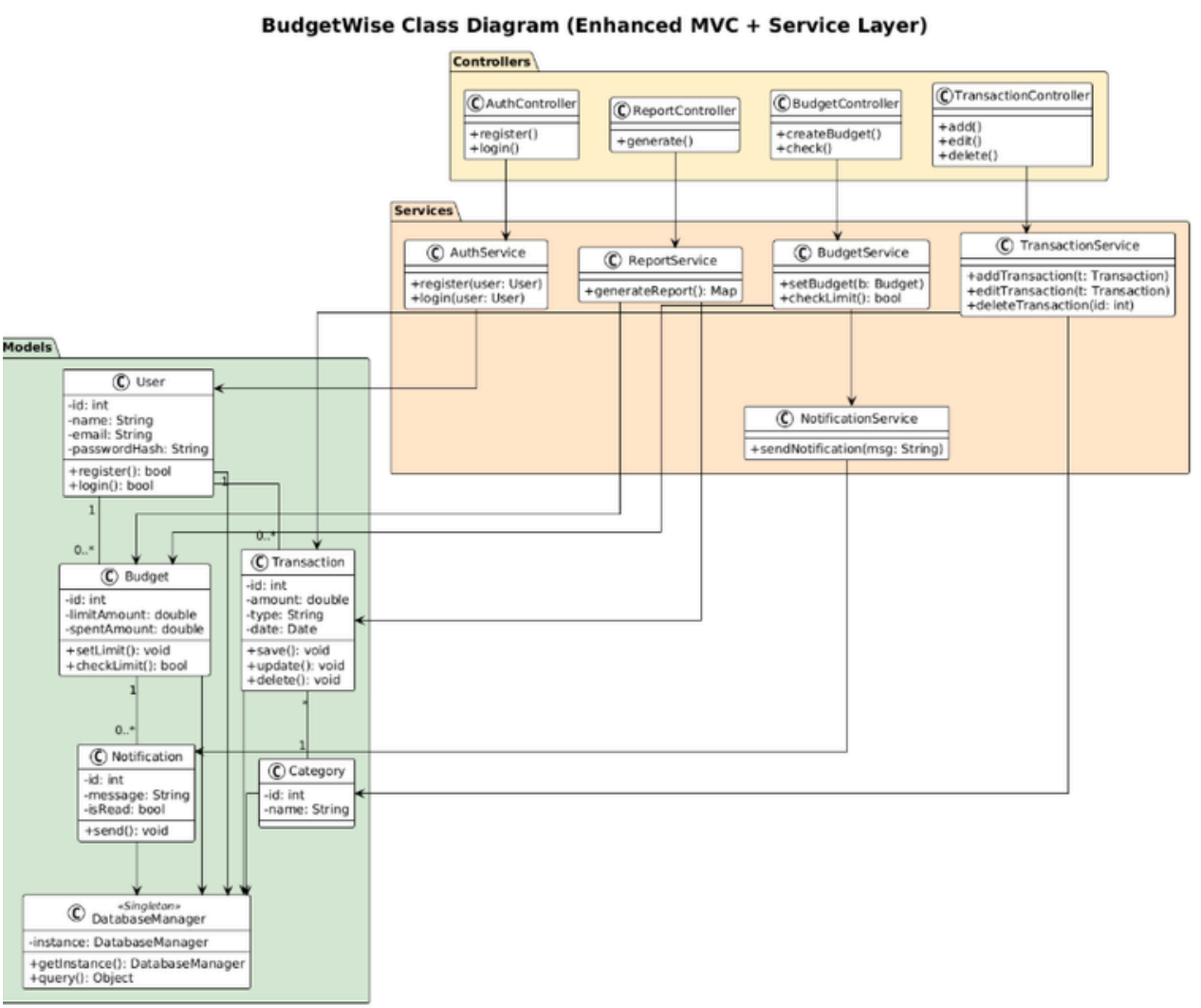


Figure A2: BudgetWise Class Diagram

PlantUML Code: See Section 2.5 — Code Block A2

2.3 Class Descriptions

ID	Class Name	Package	Description & Responsibility
C01	**User**	Model	Stores user credentials, profile info, and settings. Central entity linked to all financial data.
C02	**Transaction**	Model	Represents a single financial event (income or expense) with amount, category, date, and note.
C03	**Budget**	Model	Defines a spending limit for a specific category over a time period. Tracks usage vs. limit.
C04	**Goal**	Model	Represents a savings target with a name, target amount, deadline, and current saved amount.
C05	**Category**	Model	A classification label for transactions (e.g., Food, Transport). Can be predefined or custom.
C06	**Notification**	Model	Stores alert messages triggered by budget thresholds or goal milestones.
C07	**DatabaseManager**	Model	Singleton class that manages all DB connections and CRUD operations via SQLite/PostgreSQL.
C08	**AuthController**	Controller	Handles user registration, login validation, session management, and password hashing.
C09	**TransactionController**	Controller	Processes add/edit/delete/filter transaction requests. Triggers budget recalculations.
C10	**BudgetController**	Controller	Manages budget creation, editing, and real-time usage tracking. Sends alerts at thresholds.
C11	**GoalController**	Controller	Handles financial goal creation, contribution logging, and progress percentage calculation.
C12	**ReportController**	Controller	Aggregates transaction data and generates chart-ready datasets for the Report View.
C13	**LoginScreen**	View	Boundary class — renders the login/sign-up forms and sends credentials to AuthController.
C14	**DashboardScreen**	View	Boundary class — displays total balance, recent transactions, budget previews, and goal summaries.
C15	**TransactionScreen**	View	Boundary class — displays transaction history with filters; provides Add/Edit/Delete UI.
C16	**BudgetScreen**	View	Boundary class — shows budget cards with progress bars; provides Create/Edit budget UI.
C17	**GoalScreen**	View	Boundary class — lists all financial goals with progress; provides Add Goal / Contribute UI.
C18	**ReportScreen**	View	Boundary class — renders pie charts and bar charts using data from ReportController.

2.4 Sequence Diagrams

Seven sequence diagrams are provided, one for each of the most important user stories in the BudgetWise SRS. Each diagram shows the interaction between View, Controller, Model, and Database layers.

2.4.1 [SD-01] User Sign-Up (US #1)

Description: New user registers by providing name, email, and password. System validates, hashes the password, and creates the account.

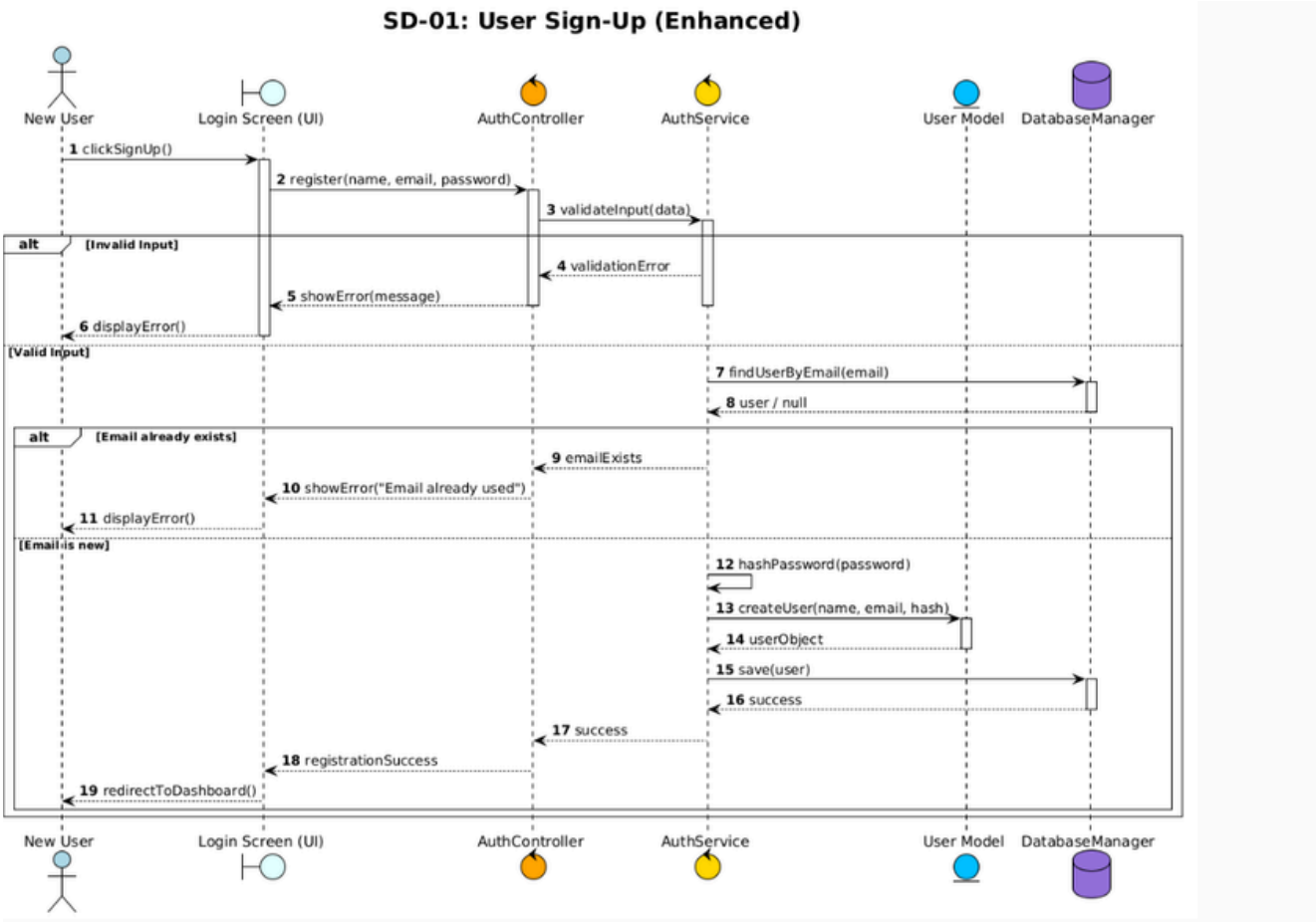


Figure SD-01: User Sign-Up (US #1)

PlantUML Code: See Section 2.5 — Code Block SD-01

2.4.2 [SD-02] User Log-In (US #2)

Description: Registered user authenticates using email and password. System verifies credentials and redirects to Dashboard.

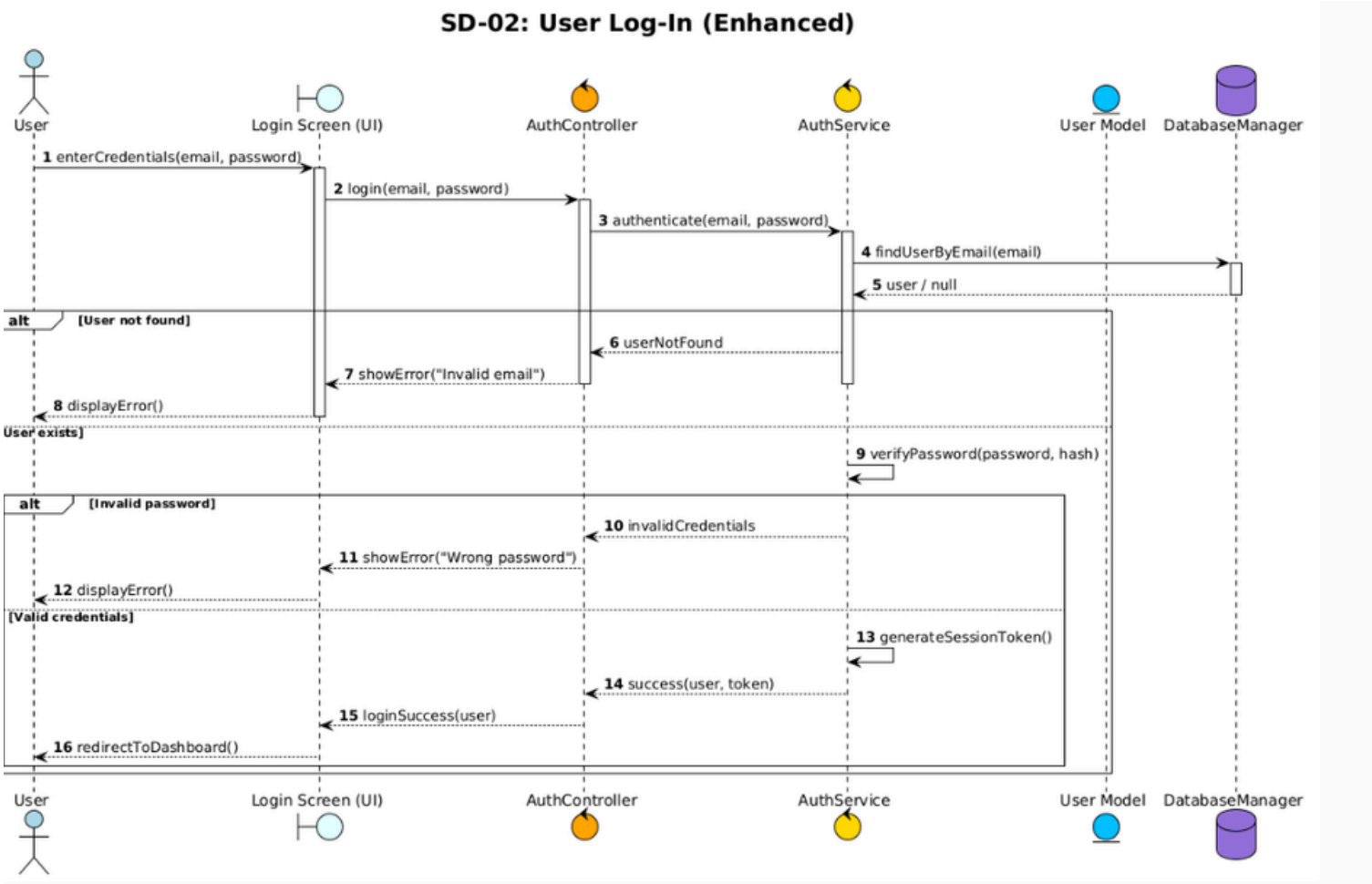


Figure SD-02: User Log-In (US #2)

PlantUML Code: See Section 2.5 — Code Block SD-02

2.4.3 [SD-03] Add Transaction (US #3)

Description: User adds a new income or expense transaction. System validates input, persists to DB, and updates Dashboard totals.

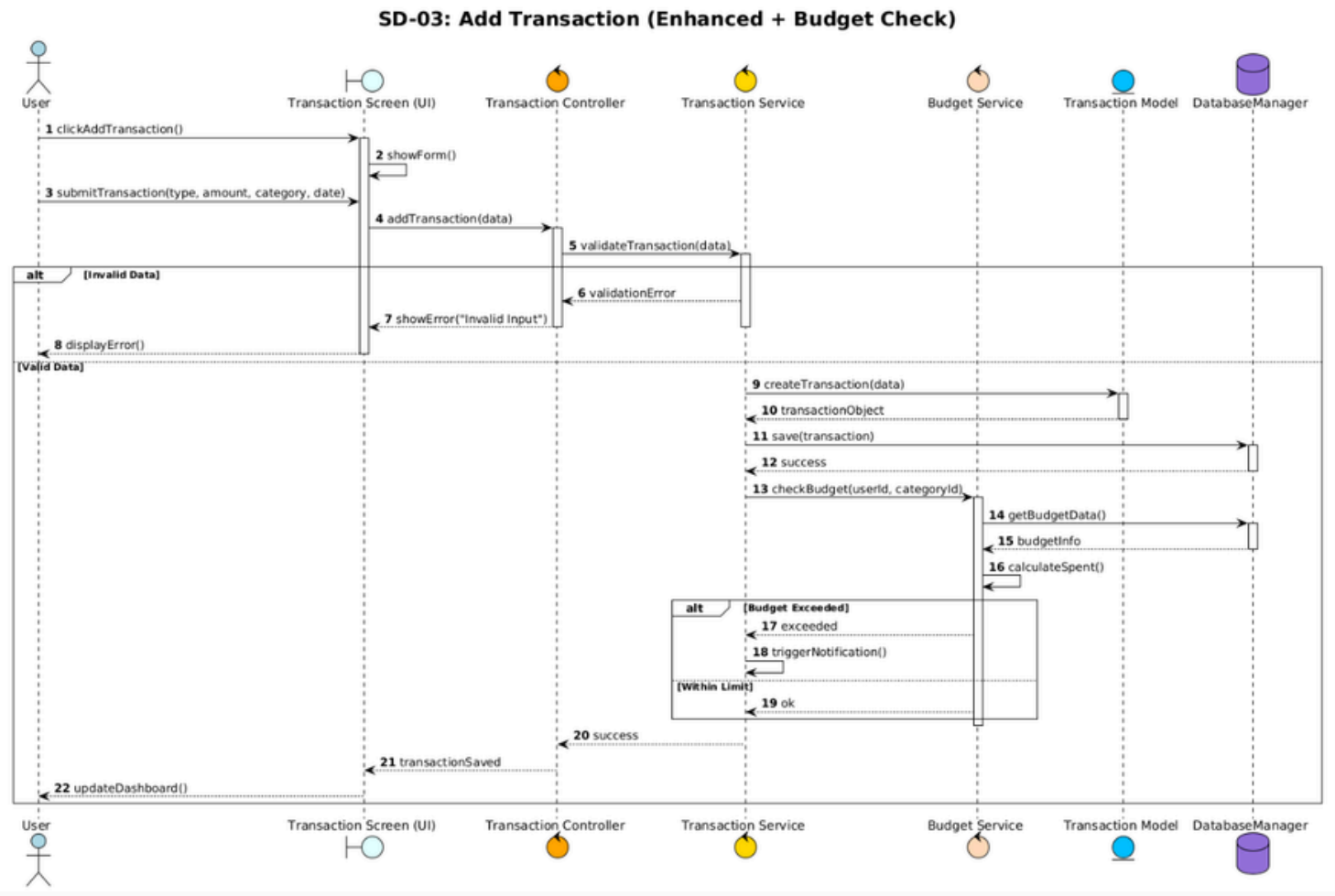


Figure SD-03: Add Transaction (US #3)

PlantUML Code: See Section 2.5 — Code Block SD-03

2.4.4 [SD-04] Create / Edit Budget (US #4)

Description: User creates a new budget for a spending category. System checks for duplicates, saves the budget, and begins tracking.

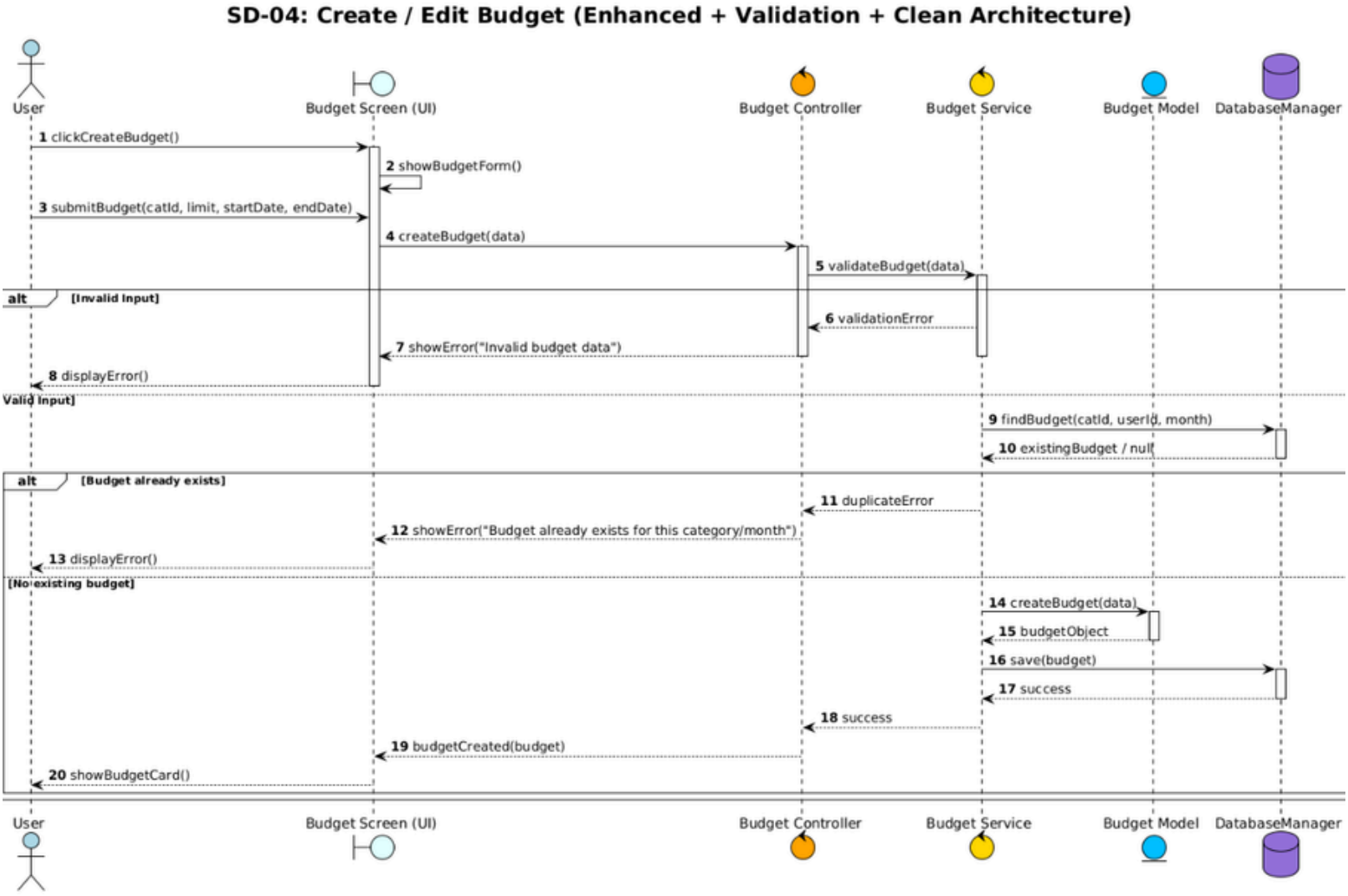


Figure SD-04: Create / Edit Budget (US #4)

PlantUML Code: See Section 2.5 — Code Block SD-04

2.4.5 [SD-05] Budget Over-Limit Alert (US #5)

Description: After logging an expense, system recalculates category spending. If threshold is reached, a notification is generated.

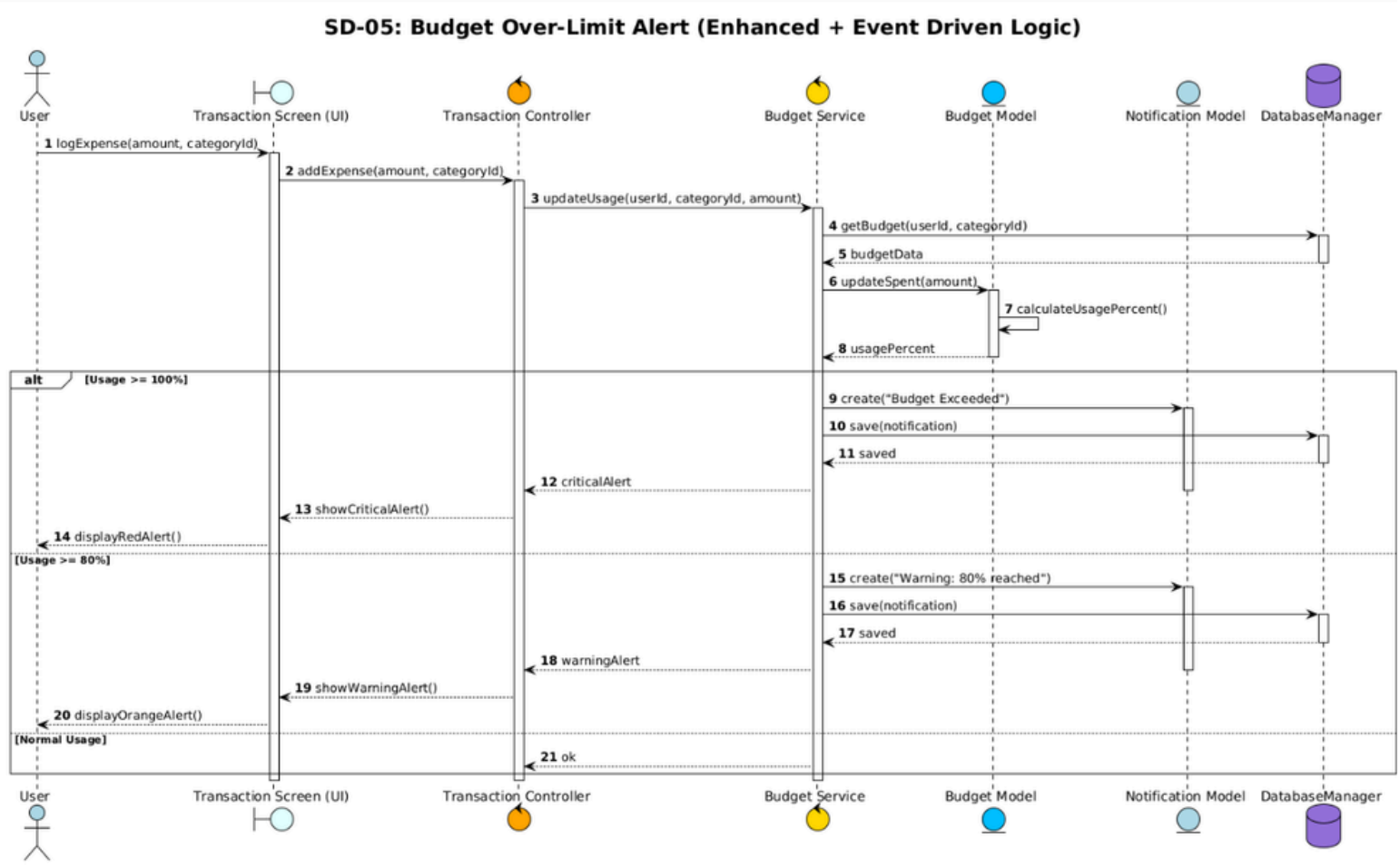


Figure SD-05: Budget Over-Limit Alert (US #5) PlantUML Code: See Section 2.5 — Code Block SD-05

2.4.6 [SD-06] Set and Track Financial Goal (US #6)

Description: User creates a savings goal with a target amount and deadline. System calculates required monthly savings and tracks progress.

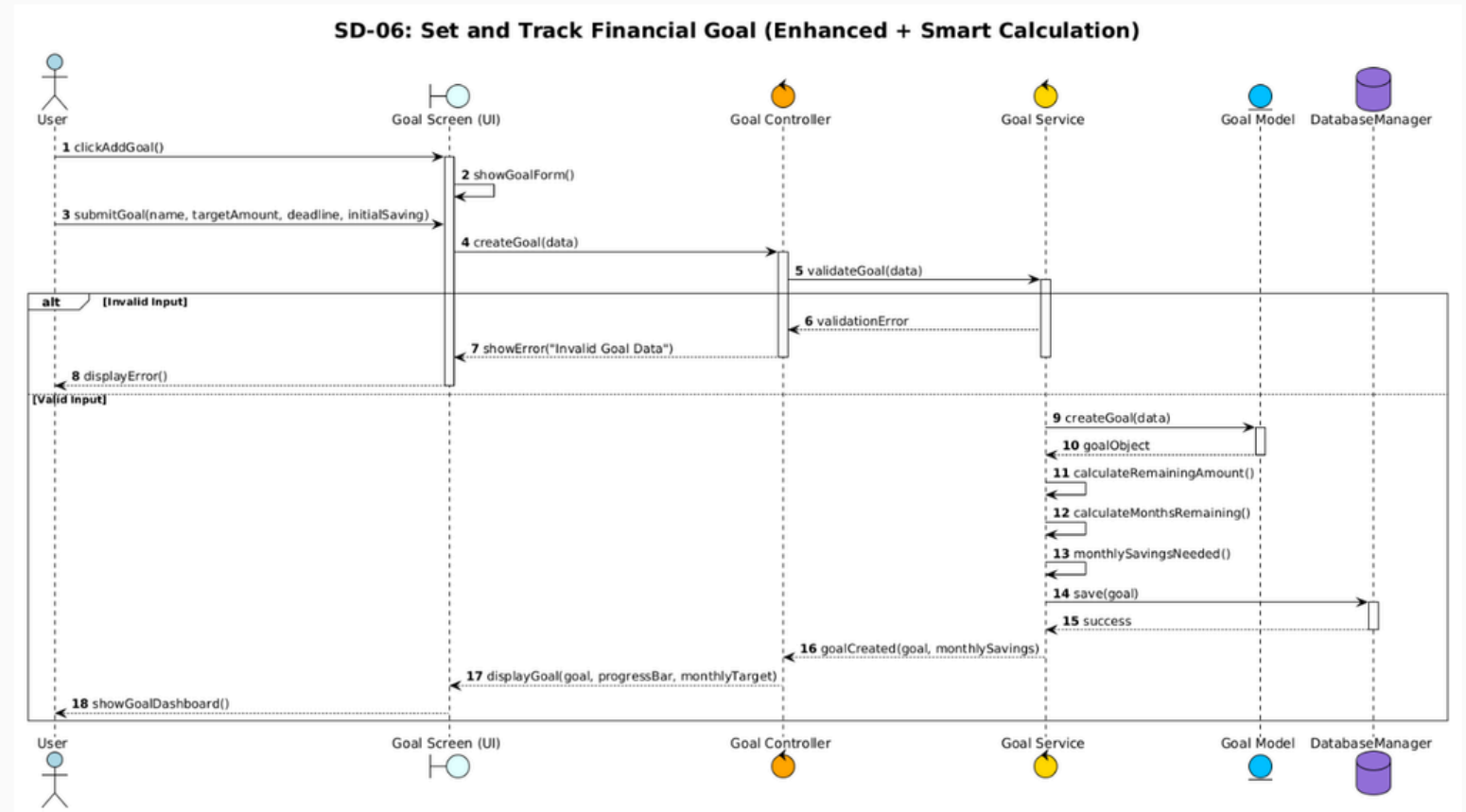


Figure SD-06: Set and Track Financial Goal

(US #6) PlantUML Code: See Section 2.5 —

Code Block SD-06

2.4.7 [SD-07] View Reports and Analytics (US #7)

Description: User navigates to Reports screen. System aggregates transaction data and renders pie and bar charts for the selected period.

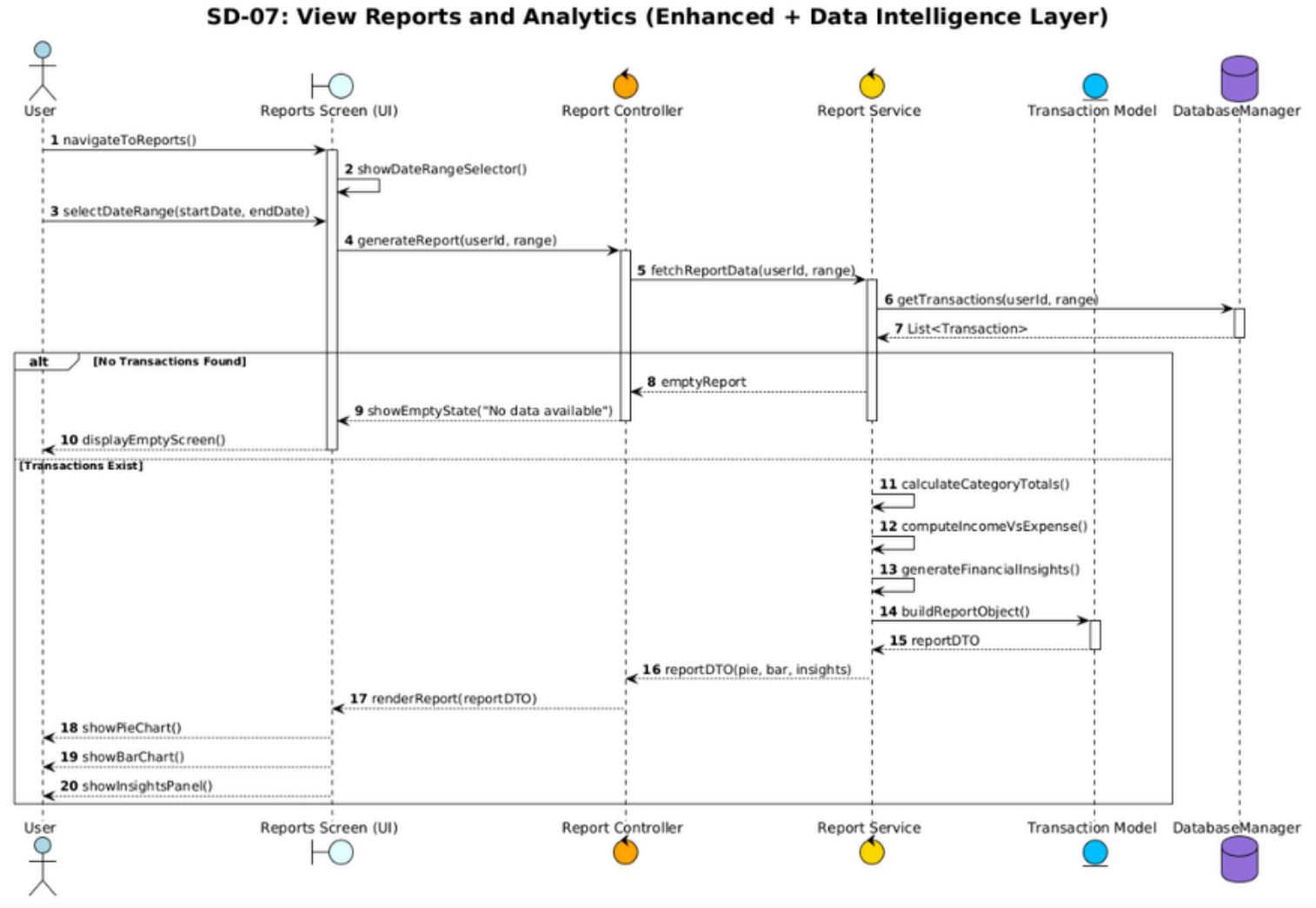


Figure SD-07: View Reports and Analytics

(US #7) PlantUML Code: See Section 2.5 —

Code Block SD-07

Class-Sequence Usage Traceability Table

Sequence Diagram	Classes Used	Key Methods
SD-01: User Sign-Up	LoginScreen, AuthController, User, DatabaseManager	register(), validateInput(), hashPassword(), save()
SD-02: User Log-In	LoginScreen, AuthController, User, DatabaseManager	login(), verifyPassword(), findByEmail()
SD-03: Add Transaction	TransactionScreen, TransactionController, Transaction, BudgetController, DatabaseManager	addTransaction(), validateInput(), checkThreshold(), save()
SD-04: Create Budget	BudgetScreen, BudgetController, Budget, DatabaseManager	createBudget(), findBudget(), save()
SD-05: Budget Alert	BudgetScreen, BudgetController, Budget, Notification, DatabaseManager	updateBudgetUsage(), getUsagePercent(), markRead(), save()
SD-06: Set Goal	GoalScreen, GoalController, Goal, DatabaseManager	createGoal(), calculateMonthlySavings(), addContribution(), save()
SD-07: View Reports	ReportScreen, ReportController, Transaction, DatabaseManager	generateReport(), aggregateCategoryTotals(), generateInsight()

Ownership Report

Item	Owner
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Class Diagram + Class Descriptions (Task 2) Sequence	Mohamed Fathy Youssef Sayed
Diagrams SD-01, SD-02, SD-03 (Task 2) Sequence	Ali
Diagrams SD-04, SD-05, SD-06, SD-07 PlantUML Code	All members
+ Class-Sequence Table	